

Grant DeWaay

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EDUCATION

Iowa State University

May 2020 – December 2024

Major: Bachelor of Science in Software Engineering

Ames, IA

Secondary Major: Interdisciplinary Design

GPA: 3.3

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Distributed Systems, Artificial Intelligence

SKILLS

Programming Languages: Java, C#, Python, F#, Golang, TypeScript, C

Libraries and Frameworks: React, .NET Core, Spring Boot, Express, Flask

Database Technologies: MySQL, MongoDB, Microsoft SQL Server, Neo4j, Chroma DB

Project Management: GitHub / GitLab, Azure DevOps, Figma, Trello

WORK EXPERIENCE

Software Engineering Intern

May 2024 – August 2024

Buildertrend

Omaha, NE

- **Developed an AI-driven rebate matching service** with C# / .NET Core that periodically scans uploaded user receipts using OCR to find eligible rebates.
- **Enhanced rebate matching accuracy by 10%** by implementing a new GPT-4o LLM pipeline, streamlining refunds for Buildertrend's users.
- **Successfully designed and proposed a new software pattern**, enabling seamless switching between LLMs in production.
- **Reduced manual validation time for testers** by implementing semantic search confidence filtering, increasing efficiency.
- **Collaborated in an Agile team** using Azure for CI/CD and project management, ensuring secure deployments with high-quality code.

Software Engineering Intern

May 2023 – August 2023

Buildertrend

Omaha, NE

- **Built a semantic search engine** in Python that takes incomplete product names and matches them with entries in Buildertrend's catalog.
- **Developed a vector search testing interface** using React that serializes search results, enabling comparisons of test sets.
- **Improved search accuracy by 30%** among internal test sets, providing a scalable foundation for future AI-driven search solutions.
- **Increased interest in AI-driven search**, encouraging Buildertrend to invest in a new development team that I joined the following summer.

PROJECTS

AI Checkers Bot

Technologies used: Java, AI-based Game Theory, Adversarial Search

- **Built a Monte Carlo Tree Search-based AI Checkers opponent** that analyzes game states to play optimal moves, making it nearly impossible to win against.
- **Developed a GUI** to visually indicate valid moves, selections, and jump sequences according.
- **Conducted performance testing** on different exploration constants to evaluate AI move quality and response, documenting the findings.

Harvard Burger – HackHarvard 2024 – Award Winner

Technologies used: Python, React, TypeScript, Docker, FFmpeg, OpenAI

- **Developed a digital drive-thru kiosk** that processes food orders by transcribing customers' verbal requests, minimizing order inaccuracies.
- **Deployed entire application** within the 36-hour event with Docker using the Defang service.
- **Won "Funniest Hack" Award** at HackHarvard.

AWARDS

HackHarvard 2024 Hackathon | Funniest Hack Award

October 2024

Iowa State College-by-College Pitch Off | 1st Place – New Idea

February 2024

Student Innovation Fund Challenge | 1st Place

May 2023

Game2Work Social Impact Game Jam | Critic's Choice Award

February 2021